OMAR FOUDA

Machine Learning Engineer | Applied AI, RAG & Computer Vision

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github.com/omarfoud | linkedin.com/in/omar-fouda-07ab01282 | kaggle.com/omaralifouda | Portfolio

PROFESSIONAL SUMMARY

Machine Learning Engineer with two years of hands-on experience building applied AI systems with Python, FastAPI, scikit-learn, and modern LLM tooling. Strong focus on retrieval-augmented generation, semantic search, computer vision, multimodal ingestion, and production API deployment. Recent work includes an Arabic-first educational RAG backend, Jobify AI matching for candidate-job recommendations, and public ML prediction APIs.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript/TypeScript, Java, Dart
ML / Deep Learning	scikit-learn, TensorFlow/Keras, PyTorch, Sentence Transformers
Applied AI / RAG	OpenAI API, Gemini API, ChromaDB, FAISS, metadata-aware retrieval, prompt workflows
Computer Vision / Media	OpenCV, Tesseract OCR, FFmpeg, Whisper transcription
Backend / Deployment	FastAPI, Docker, WebSockets, Uvicorn, Streamlit, ngrok, REST APIs
Data / Databases	NumPy, Pandas, Matplotlib, Seaborn, SQL Server, PostgreSQL, MySQL, MongoDB, SQLite
Tools	Git/GitHub, Jupyter, Google Colab, VS Code, Kaggle

SELECTED PROJECTS

Educational RAG Backend

Python | FastAPI | ChromaDB | OpenAI/Gemini | OCR | Transcription | Docker

https://github.com/omarfoud/Educational_rag

- Built an Arabic-first educational RAG backend for documents, images, audio, and video.
- Designed multimodal ingestion with FFmpeg, OCR, and transcription workflows to extract searchable learning content.
- Implemented metadata-aware retrieval across subject, grade, term, book, and course-book context for more relevant answers.
- Prepared local and low-cost cloud deployment profiles with Docker and provider switches for embeddings, OCR, and transcription.

Jobify AI Matching Backend - Graduation Project

Python | FastAPI | FAISS | Sentence Transformers | Gemini | SQL Server | SQLite

https://github.com/omarfoud/gradution_project

- Built an AI backend for job recommendations, CV analysis, and career chat connected to a hosted SQL Server application database.
- Combined FAISS semantic search with SQLite/SQL fallback and active CV retrieval for candidate-job matching workflows.
- Implemented PDF/DOCX CV text extraction, embeddings, Gemini gap analysis, match percentages, missing skills, and improvement advice.
- Added health checks and admin sync endpoints to keep job and resume embeddings current in production workflows.

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ADDITIONAL PROJECT

Diabetes Prediction Public API

Python | FastAPI | scikit-learn | ngrok | REST API

https://github.com/omarfoud/diabetes-public-api

- Exposed a trained scikit-learn diabetes prediction model as a public REST API using FastAPI and ngrok.
- Created a JSON request/response contract for eight clinical inputs and returned human-readable prediction results.
- Packaged the deployment flow with a serialized model, FastAPI wrapper, ngrok tunnel startup, and requests-based endpoint tests.

EDUCATION

Bachelor of Science - Computers and Information (Information Systems)

Faculty of Computers and Information, Menoufia University | 2022 - 2026

CERTIFICATIONS

- Kaggle - Computer Vision | Nov 2025
- Kaggle - Intro to Deep Learning | Nov 2025
- Information Technology Institute (ITI) - Full Stack Web Development using Python, 160 hours | Jul-Aug 2025
- Manara - Responsible AI and Ethical Considerations | Jun 2024

LANGUAGES

Arabic: Native | English: Professional working proficiency

CURRENT FOCUS

- Computer vision with OpenCV: image processing, feature detection, and real-time video analysis.
- LLM integration and RAG pipelines for Arabic and multilingual educational content.
- Production API deployment with FastAPI, Docker, WebSockets, and cloud-friendly provider profiles.
- Open-source machine learning projects and Kaggle practice across prediction, classification, and recommendation tasks.